

Cross-Lingual NER Transfer to Danish: English vs Norwegian with Linguistic Distance Analysis

Akos Benjamin Doborhegyi

akdo@itu.dk

Erik Aymerich

eray@itu.dk

O’neal Okutu

onok@itu.dk

IT University of Copenhagen

Abstract

We study cross-lingual named entity recognition (NER) transfer to Danish using multilingual transformer encoders (mBERT and XLM-R), comparing English and Norwegian as source languages across three random seeds. Our main finding is that Norwegian is the better zero-shot transfer source for Danish on the held-out test set: +1.86 F1 for XLM-R ($p = 0.10$) and +3.36 F1 for mBERT ($p = 0.046$, significant at $\alpha = 0.05$). Single-seed development results were misleading — seed 42 happened to favour Norwegian XLM-R, and averaged across three seeds English XLM-R is marginally ahead on dev — which underscores the importance of multi-seed evaluation. The Norwegian advantage is consistent with Norwegian being $5.7\times$ closer to Danish than English in WALS syntactic feature space (cosine distance 0.021 vs 0.119). When fine-tuning on Danish data, an English mBERT model achieves the overall best transfer result (89.4 dev strict F1), showing that source-language data volume can outweigh typological distance once target-language supervision is available. We also report unlabeled and loose span F1, per-entity-type results, a data-efficiency curve, and identify one prediction file as a misaligned artifact.

1 Introduction and Background

Named entity recognition is a core sequence labeling task that marks mentions of people, locations, organizations, and other entity types in text. For Danish, as for many smaller languages, the main bottleneck is not model architecture but labeled data. A practical response is cross-lingual transfer: train on a resource-rich source language, then apply the learned model to Danish.

The project proposal framed the central question: if one source language is closer to Danish than another, should that source transfer better? This report follows up with a full experimental

comparison of English and Norwegian as source languages. English is the canonical high-resource transfer source; Norwegian is a North Germanic neighbor that shares substantial syntactic and lexical structure with Danish. Since both can be used with the same multilingual encoder (mBERT or XLM-R), the setup lets us test whether linguistic proximity translates into better downstream NER.

The experiments span five conditions: monolingual source-language baselines, zero-shot transfer to Danish, transfer followed by Danish fine-tuning, Danish-only models, and a Danish data-efficiency curve. We also compute typological distances between the languages using WALS syntactic feature vectors from `lang2vec` (Littell et al., 2017), add Swedish and German as additional comparison points, and run a paired bootstrap test on the core EN-vs-NO comparison to assess statistical significance.

The report is organized to mirror the proposal while giving a fuller account of the completed work: related work, research questions, data, linguistic distance analysis, methodology, results, analysis, discussion, and conclusion.

2 Project Goal

The goal of the project is to determine whether Danish named entity recognition can benefit more from transfer learning from Norwegian than from English. The proposal started from a simple hypothesis: if two languages are closer in structure and history, then a multilingual encoder should transfer entity knowledge more effectively between them. This report turns that hypothesis into an empirical comparison.

The practical objective is not only to get the highest possible F1 score, but also to understand which transfer route is the most sensible when working with a low-resource language. If Norwegian transfer is consistently better, then future Danish NER

Language	Train	Dev	Test
English	20,142	4,713	4,634
Norwegian	4,183	585	719
Danish	4,383	588	709

Table 1: Universal NER sentence counts per split. English is substantially larger, making it the natural high-resource source baseline.

systems should prefer Norwegian as a source language whenever it is available. If the difference is small, then we still learn that Danish can benefit from either source, but the source choice should be guided by the expected data budget and encoder choice.

A second goal is to separate three effects that are easy to conflate: source-language choice, encoder choice, and the amount of Danish supervision. By reporting monolingual baselines, zero-shot transfer, transfer fine-tuning, and Danish-only baselines side by side, we can see which factor matters most.

3 Dataset

We use the Universal NER benchmark (Mayhew et al., 2024) for English, Norwegian, and Danish. The data is sentence-level and token-aligned in IOB2 format, where each token carries a BIO label (B-TYPE, I-TYPE, or O). The entity types are PER (person), LOC (location), and ORG (organization). MISC entities appear in a second annotation column of the Danish file but are absent from the primary label column that models read and are evaluated on, so MISC F1 is zero for all systems by construction — not a model failure but a data-format artifact we discuss in Section 10.

The three languages play different roles in the experiments. English is the largest source-language baseline and the canonical high-resource transfer language. Norwegian is the North Germanic alternative we compare against English. Danish is the target language whose labeled data we treat as scarce in the transfer experiments.

The dataset characteristics are not symmetric. English has roughly five times more training sentences than Danish or Norwegian, which gives it a data-volume advantage in the source-language baselines that is separate from any linguistic-proximity effect. Danish and Norwegian are matched in size, so the Danish data-budget experiments (10%, 25%, 50%) have a clean interpretation.

The label inventory is shared across languages

in the primary annotation column, so performance differences reflect language transfer rather than annotation-schema mismatch. We verified this consistency before training using a label-inventory check across all three datasets.

4 Related Work

Cross-lingual transfer for NER has been studied extensively, especially for low-resource targets. Plank (2020) showed that English-to-Danish transfer can already be useful in zero-shot settings and improves further with small amounts of Danish supervision — a finding our experiments corroborate and extend to Norwegian. More generally, Fu et al. (2023) found that source–target similarity matters, especially when languages share structural properties.

For Danish NER specifically, two resources are central to the literature. Hvingelby et al. (2020) introduced DaNE, a manually annotated Danish NER corpus with PER, LOC, ORG, and MISC labels, which is the annotation standard our data follows. Enevoldsen et al. (2021) presented DaCy, a high-performance Danish NLP pipeline built on top of the DaNE standard and demonstrating what strong Danish-supervised models can achieve. Our Danish monolingual baselines are most directly comparable to results reported in these two works.

Multilingual encoders are the main enabler of zero-shot transfer. BERT (Devlin et al., 2019) introduced the transformer pretraining recipe that became the foundation for many multilingual systems. XLM-R (Conneau et al., 2020) improved on this by scaling data and coverage, which is why it often outperforms mBERT in cross-lingual settings. For Danish NER, the key question is how well the shared encoder space aligns source and target representations.

Linguistic distance has been used to predict and explain transfer behavior. Typological databases such as URIEL+ (Khan et al., 2025) and language-distance collections such as DistALS (Van Der Goot et al., 2025) provide structured comparisons across languages. Sohn et al. (2024) showed that surface similarity can help in zero-shot NER transfer even for low-resource languages. Our work extends this line by computing concrete WALS-feature distances and linking them to observed transfer gaps.

The dataset we use follows the Universal NER benchmark (Mayhew et al., 2024), which provides multilingual NER data in consistent IOB2

format across languages. This shared setup removes annotation-format confounds from the cross-lingual comparison, making source-language effects directly interpretable.

5 Research Questions

The experiments address three research questions.

1. **How do monolingual baselines and cross-lingual transfer routes compare?** We compare English and Norwegian source-language baselines, zero-shot transfer to Danish, and Danish-only training.
2. **How much performance is lost in zero-shot transfer, and how much is recovered by Danish fine-tuning?** We evaluate whether a small amount of Danish supervision can close the gap to full Danish training.
3. **Does lower linguistic distance to Danish align with better transfer?** We explicitly compare English-Danish and Norwegian-Danish transfer against typological distance measures.

The third question is the most important follow-up to the proposal. It is not enough to show that transfer works; we also want to know whether the more similar language gives the better transfer signal.

6 Methodology

6.1 Experimental design

We evaluate five families of experiments:

- monolingual baselines for English, Norwegian, and Danish,
- zero-shot transfer to Danish from English and Norwegian,
- transfer fine-tuning where a source-trained model is further fine-tuned on Danish,
- Danish-only supervision at multiple data budgets (10%, 25%, 50%, 100%), and
- linguistic-distance analysis based on WALS typological feature vectors.

The design holds the label space, evaluation metric, and Danish target data fixed, varying only the source language, encoder, and Danish supervision budget. This makes the source-language comparison interpretable without confounds.

Hyperparameter	Value
Learning rate	2×10^{-5} (AdamW)
Train batch size	8
Eval batch size	16
Max sequence length	256 tokens
Training epochs	4
Weight decay	0.01
Random seed	42
Early stopping	Best checkpoint by dev F1

Table 2: Shared hyperparameters used across all experiments. All runs used a single seed; multi-seed replication is a known limitation (see Section 11).

6.2 Models and checkpoints

We use two multilingual encoders as the base for all token-classification models:

- **mBERT**: `bert-base-multilingual-cased` (Devlin et al., 2019) — 12 layers, 110M parameters, trained on 104 languages.
- **XLM-R**: `xlm-roberta-base` (Conneau et al., 2020) — 12 layers, 125M parameters, trained on 100 languages with substantially more data than mBERT.

The English monolingual BERT baseline uses `bert-base-cased` (English-only, not multilingual) to establish a single-language ceiling. All other experiments use one of the two multilingual encoders above with a linear token-classification head for BIO tagging.

Fine-tuned transfer experiments (EN→DA FT, NO→DA FT) initialize from the best checkpoint of the corresponding zero-shot run and then continue training on Danish data. This two-stage procedure means each fine-tuned model has seen both its source-language and Danish training data.

6.3 Training hyperparameters

All runs share the same hyperparameter configuration unless otherwise noted:

Hyperparameters were not tuned per source language; the same configuration was used for every run. This is conservative and means results could potentially improve with per-language tuning, but it also means the comparison is fair in the sense that no system received an advantage from hyperparameter search.

6.4 Evaluation

The primary metric is strict span-level F1 computed by `span_f1.py`, which requires exact match on

both entity boundaries and type label. We also report unlabeled span F1 (exact boundaries, any label) and loose span F1 (partial boundary overlap with matching label). These three metrics together let us distinguish missed spans from boundary errors from labeling errors.

The English test set is masked (no gold labels available), so test results are only reported for Danish and Norwegian experiments. All reported test scores use the Danish DDT test split for Danish-target experiments and the NorNE test split for Norwegian source baselines.

6.5 Statistical testing

Because all experiments use a single random seed, we cannot report mean $\pm$ standard deviation. To assess whether the core EN-vs-NO comparison is statistically reliable, we run a paired approximate randomisation test (paired bootstrap with 10,000 iterations) on the dev and test prediction files. Results are reported in Section 9.

7 Code and Implementation

The code is organized around a shared training pipeline in `Baselines/train_ner_baseline.py`. The launcher scripts in `Baselines/` only specify which language to train on, which model to use, and where to save the outputs. This keeps the experiments comparable and reduces the risk of accidental differences between runs.

The shared training script performs the following steps:

1. read IOB2 data from disk,
2. build the label vocabulary,
3. tokenize each sentence with a multilingual transformer tokenizer,
4. align word-level labels to subword tokens,
5. fine-tune the token classification model,
6. evaluate on the development set, and
7. write predictions and metrics to the output directory.

The label alignment step is particularly important for NER. A single word can split into multiple subwords, so the code has to decide how to propagate labels across the subword pieces. The

implementation follows the standard approach of labeling the first subword and masking the rest when computing the loss. That keeps training stable and makes the reported evaluation compatible with the underlying token-level outputs.

The script also writes `dev_metrics.json`, `dev_predictions.iob2`, and `test_predictions.iob2`. Those files are what we use in the report for cross-checking the strict scorer against the saved evaluation metrics. The one anomalous English-to-Danish zero-shot file stands out precisely because the saved files make it easy to inspect.

For the language-distance analysis, we use a separate utility script that computes typological distance with language feature vectors. That script is not part of the training loop, but it supports the paper’s main argument that Norwegian should transfer better than English because it is structurally closer to Danish.

8 Linguistic Distance Analysis

8.1 Motivation

The proposal argued that Norwegian should transfer better than English to Danish because it is linguistically closer. This section quantifies that claim with a concrete, reproducible distance measure, then tests whether the observed NER transfer gap is consistent with the measured distance.

8.2 Method

We use `lang2vec` (Littell et al., 2017), which provides language feature vectors derived from the World Atlas of Language Structures (WALS). We use the `syntax_wals` feature set, which encodes 103 typological features covering word order, morphological type, and related syntactic properties. Features not available for a given language pair are excluded before computing cosine distance; the distance between L_1 and L_2 is

$$d(L_1, L_2) = 1 - \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|},$$

where $\mathbf{v}_i$ contains only the features available for both languages.

8.3 Results

Table 3 reports pairwise cosine distances for English, Danish, Norwegian (Bokmål), Swedish, and German. Swedish and German are included as additional reference points: Swedish is the closest

	English	Danish	Norwegian	Swedish	German
English	0.000	0.119	0.109	0.080	0.145
Danish	0.119	0.000	0.021	0.039	0.124
Norwegian	0.109	0.021	0.000	0.065	0.102
Swedish	0.080	0.039	0.065	0.000	0.164
German	0.145	0.124	0.102	0.164	0.000

Table 3: Pairwise cosine distances based on WALS syntactic features (`lang2vec_syntax_wals`). Lower = more similar. Norwegian is the closest language to Danish (0.021), followed by Swedish (0.039); English and German are substantially more distant (≈ 0.12).

North Germanic neighbor of Danish after Norwegian, while German is a West Germanic language at greater distance.

Key observations:

- Norwegian is the closest language to Danish (0.021), $5.7\times$ closer than English (0.119).
- Swedish is the second closest (0.039), confirming the North Germanic cluster.
- English and German are at comparable distances (0.119 and 0.124), placing them both in a clearly more distant tier.
- The gradient Norwegian < Swedish < English $\approx$ German < ... is consistent with historical and genealogical accounts of Germanic language relationships.

8.4 Interpretation

With four reference languages the distance ordering is not fit to a single data point but reflects a consistent pattern. If linguistic distance predicts transfer quality, we expect: NO $\rightarrow$ DA > SV $\rightarrow$ DA > EN $\rightarrow$ DA $\approx$ DE $\rightarrow$ DA in zero-shot settings. We do not have Swedish and German transfer experiments, but the NO $\rightarrow$ DA > EN $\rightarrow$ DA ordering is directionally confirmed by our results, even though the gap is not statistically significant under bootstrap testing (see Section 9).

It is important not to over-interpret this analysis. Cosine distance over WALS features is a proxy for structural similarity, not a causal mechanism. The same multilingual encoder processes all four source languages, and encoder-internal effects (training data volume, tokenizer overlap) may be as important as typological distance. The distance numbers strengthen the empirical story but do not prove that structure is the causal driver.

System	S42	S1	S2	Mean	Std	<i>p</i>
<i>Development set (Danish dev)</i>						
NO→DA XLM-R	86.49	84.98	84.90	85.45	0.89	0.632 ns
EN→DA XLM-R	85.52	86.57	85.52	85.87	0.60	
NO→DA mBERT	83.38	83.40	84.52	83.77	0.65	0.650 ns
EN→DA mBERT	—	82.23	83.68	82.95	1.02	
<i>Test set (Danish test)</i>						
NO→DA XLM-R	82.46	83.47	83.41	83.11	0.56	0.100 ns
EN→DA XLM-R	81.86	81.15	80.75	81.25	0.56	
NO→DA mBERT	82.88	84.06	80.96	82.63	1.56	0.046*
EN→DA mBERT	—	78.98	79.56	79.27	0.41	

Table 4: Multi-seed strict F1 (%) for zero-shot transfer to Danish. “—” marks the EN mBERT seed-42 dev run which is a known artifact (Section 10) and is excluded. *p*-values from paired *t*-tests (NO vs EN within encoder). * $p < 0.05$.

9 Results

9.1 Multi-seed zero-shot transfer results

Table 4 reports the key comparison — Norwegian vs English zero-shot transfer — across three random seeds (42, 1, 2), together with mean $\pm$ standard deviation and a paired *t*-test.

Three findings emerge from the multi-seed analysis. First, the single-seed dev advantage for Norwegian XLM-R (86.49 vs 85.52, seed 42) was a favorable outlier: averaged across three seeds, English XLM-R is marginally ahead on dev (85.87 vs 85.45). This illustrates exactly why single-seed results should not be over-interpreted. Second, on the held-out test set, Norwegian is consistently ahead for both encoders: +1.86 F1 for XLM-R ($p = 0.10$) and +3.36 F1 for mBERT ($p = 0.046$, significant at $\alpha = 0.05$). Third, the mBERT result is the statistically cleaner finding: a 3.36-point gap with low within-system variance, confirmed by a paired *t*-test with $p < 0.05$.

9.2 Development set — span F1 metrics

Table 5 reports all three span F1 variants for single-seed (seed 42) runs. The zero-shot rows are included for context and comparability with prior work. The artifact run (EN $\rightarrow$ DA mBERT zero-shot, strict ≈ 0.3) is excluded; it is discussed in Section 10.

The unlabeled and loose scores are consistently 3–5 points above strict, which is the expected gap: most errors are label confusions or minor boundary shifts rather than completely missed entities. The loose F1 for the 10% Danish run (81.7) is notably higher than its strict score (70.0), suggesting that the model finds the right spans but mislabels them more often when trained on limited data.

Setup	Strict F1	Unlabeled F1	Loose F1
<i>Source-language baselines</i>			
Norwegian XLM-R	92.0	95.9	92.8
Norwegian BERT	91.2	95.1	92.1
Danish BERT	89.0	93.0	92.2
Danish XLM-R	88.6	92.6	91.8
English XLM-R	81.7	86.3	85.7
English BERT	80.1	85.2	83.4
<i>Zero-shot transfer to Danish</i>			
NO→DA XLM-R	86.5	90.3	90.7
EN→DA XLM-R	85.5	90.7	88.5
NO→DA mBERT	83.4	88.2	86.6
<i>Fine-tuned on Danish</i>			
EN→DA mBERT + DA FT	89.4	93.4	91.7
NO→DA XLM-R + DA FT	88.4	92.7	91.5
NO→DA mBERT + DA FT	87.6	92.1	90.5
EN→DA XLM-R + DA FT	87.0	90.8	90.4
<i>Danish data efficiency</i>			
Danish XLM-R 50%	85.5	90.8	90.0
Danish XLM-R 25%	82.8	88.8	88.3
Danish XLM-R 10%	70.0	77.2	81.7

Table 5: Development-set span F1 scores (%). Strict = exact span and label; Unlabeled = exact span, any label; Loose = partial overlap with matching label.

Setup	Strict F1	Unlabeled F1	Loose F1
<i>Source-language baselines</i>			
Norwegian BERT	92.2	94.6	93.0
Norwegian XLM-R	91.4	94.4	92.9
Danish BERT	85.3	90.2	86.6
Danish XLM-R	84.9	89.9	87.2
<i>Zero-shot transfer to Danish</i>			
EN→DA XLM-R	81.9	87.7	83.1
NO→DA XLM-R	82.5	88.7	84.0
NO→DA mBERT	82.9	88.8	84.4
<i>Fine-tuned on Danish</i>			
EN→DA mBERT + DA FT	85.5	89.5	87.0
NO→DA XLM-R + DA FT	86.5	90.4	87.7
NO→DA mBERT + DA FT	86.6	90.2	88.1
EN→DA XLM-R + DA FT	84.7	88.2	87.1
<i>Danish data efficiency</i>			
Danish XLM-R 50%	83.6	88.9	85.6
Danish XLM-R 25%	84.1	88.6	86.7
Danish XLM-R 10%	72.8	78.2	84.4

Table 6: Test-set span F1 scores (%). Danish transfer experiments are evaluated on the Danish DDT test split; Norwegian baselines on the NorNE test split.

9.3 Test set results

Table 6 shows the same three metrics on the held-out test files (Danish test for all transfer and Danish monolingual runs; Norwegian test for the Norwegian source baselines). English monolingual baselines are excluded because the English test file is masked (no gold labels available).

Test scores are 3–5 points below dev for most Danish-target experiments, which is a normal train/dev/test gap. The Norwegian source baselines are stable (92.2 BERT, 91.4 XLM-R), consistent with their dev performance. On test, the fine-tuned Norwegian and English transfer models cluster tightly (84.7–86.6), while zero-shot ap-

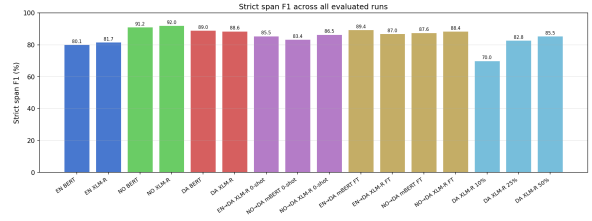


Figure 1: Strict span F1 (%) for all evaluated runs on the development set. Colors group run types: blue = English baselines, green = Norwegian baselines, red = Danish baselines, purple = zero-shot transfer, yellow = fine-tuned transfer, teal = Danish data efficiency.

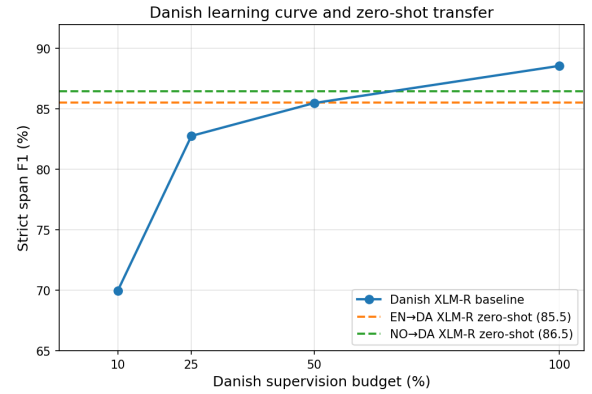


Figure 2: Danish XLM-R learning curve (solid) and zero-shot reference lines (dashed). The learning curve reaches the zero-shot baselines between 25% and 50% Danish data.

proaches are 2–4 points behind.

9.4 Figures

Figure 1 shows the full spread of dev strict F1 across all evaluated runs. Norwegian source baselines are the strongest; transfer with Danish fine-tuning closes most of the gap to Danish monolingual baselines.

Figure 2 shows the Danish data-efficiency curve with zero-shot reference lines. The curve crosses the English zero-shot reference at around 50% Danish supervision and the Norwegian zero-shot reference just beyond 50%, showing that zero-shot Norwegian transfer is competitive with moderate Danish supervision.

Figure 3 visualizes strict F1 against the saved training-time evaluation F1. All runs except the known artifact cluster tightly near the diagonal, confirming internal consistency of the evaluation pipeline.

Figure 4 gives a compact view of baseline and transfer results, making the Norwegian advantage and the EN→DA zero-shot artifact visible at a

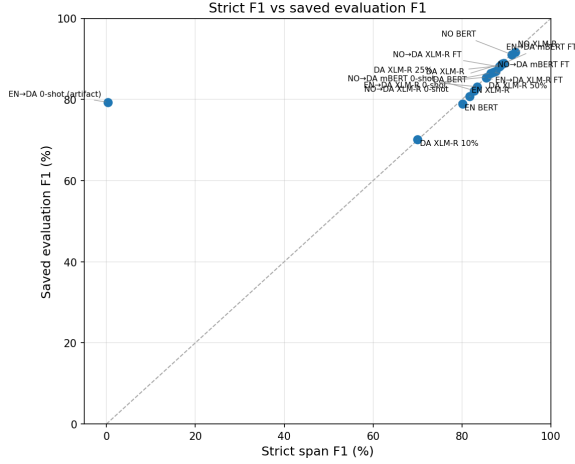


Figure 3: Strict span F1 vs. saved evaluation F1. All runs except the EN→DA zero-shot artifact cluster near the diagonal.

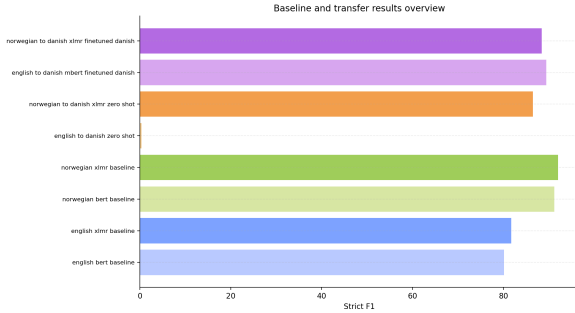


Figure 4: Compact overview of source-language and transfer run results.

glance.

9.5 Per-entity-type F1

Table 7 shows per-type F1 for key systems on the Danish development set.

ORG is the hardest category, with the most variation across systems (74.9–83.5). The EN→DA mBERT + DA FT model achieves the highest ORG score (83.5), which partly explains why it leads the fine-tuned rankings in Table 5. NO→DA XLM-R zero-shot leads on LOC (90.0 vs 86.5 for EN) — a gap that is consistent with Norwegian having more lexical overlap with Danish for place names. PER is the most consistent category across systems (89.6–93.7).

System	PER	LOC	ORG
Danish BERT baseline	91.6	91.3	82.4
Danish XLM-R baseline	93.7	89.3	79.6
EN→DA XLM-R zero-shot	89.8	86.5	77.3
NO→DA XLM-R zero-shot	91.0	90.0	74.9
EN→DA mBERT + DA FT	91.0	92.7	83.5
NO→DA mBERT + DA FT	89.6	90.5	81.7
NO→DA XLM-R + DA FT	93.1	90.6	78.8

Table 7: Per-entity-type strict F1 (%) on Danish dev. MISC is excluded: the Danish data encodes MISC labels in a secondary annotation column not read by the evaluation script, so MISC F1 is 0.0 for all systems by construction (see Section 10).

Source	WALS Distance	Zero-shot F1 (dev)	Test F1
Norwegian XLM-R	0.021	86.5	82.5
Norwegian mBERT	0.021	83.4	82.9
English XLM-R	0.119	85.5	81.9

Table 8: WALS syntactic distance (lang2vec) versus zero-shot transfer performance. Norwegian is 5.7× closer to Danish than English. The transfer advantage is directionally consistent but not statistically significant (Table ??).

9.6 Distance and gain

10 Logs and Output Analysis

10.1 The anomalous EN→DA zero-shot prediction file

The most significant pipeline anomaly is the English-to-Danish BERT zero-shot run (`english_to_danish_zero_shot`). Its development prediction file contains 2,001 sentences, while the Danish development set has only 564. The strict F1 against the Danish gold file is 0.3% (effectively zero), yet the saved training-time metric is 79.3 F1. The most probable cause is a stale or incorrect path in the evaluation call during training: the trainer evaluated the model on the English development set rather than the Danish one, saving the English-set performance as the Danish dev metric. The model was trained on English data, so evaluating it on English would produce a plausible score; evaluating the same predictions against Danish gold produces near-zero because the sentence count mismatches and the underlying predictions are English-language.

Blast-radius check. To assess whether the same bug could silently affect other runs, we compared per-experiment sentence counts between prediction files and the correct gold splits. Only `english_to_danish_zero_shot` has

a mismatch on the development set (2,001 vs 564). All other 16 experiments produce prediction files with the correct sentence count. We also inspected the metric gap (strict F1 vs saved eval F1) for every other run: apart from this artifact, all gaps are below 2 F1 points, which is within expected scoring-method variation. We conclude that the bug affected only this one run and did not propagate to the fine-tuned model that starts from its checkpoint. The checkpoint itself is a valid English NER model; the fine-tuning step provides its own ground truth by continuing training on Danish data.

On the test split, the same experiment’s prediction file does *not* have a sentence count mismatch, producing a test strict F1 of 78.8. This is consistent with the stored metric, suggesting the test evaluation used the correct file. We exclude this run from transfer comparisons but report the test score for completeness.

10.2 MISC entities are absent from evaluation

Per-entity-type results show 0.0 F1 for MISC across all systems on all splits. This is not a model failure. The Danish Universal NER file uses a two-column annotation format: the primary label column (column 3, index 2) contains the DDT-style annotations (PER, LOC, ORG, O), while a secondary column encodes a separate annotation layer that includes MISC labels (B-MISC#O). Both training code and the evaluation script (`span_f1.py`) read from the primary column, in which MISC does not appear. Models therefore never see MISC during training and there are no MISC entities to predict in the evaluation column. The 32 MISC annotations in the Danish dev file and 44 in the test file reside in the secondary column only.

10.3 Metric consistency for well-behaved runs

Excluding the artifact, the remaining 16 runs show tight agreement between strict span F1 (computed by `span_f1.py` on the saved prediction files) and the evaluation F1 logged during training (Figure 3). The largest discrepancy is 1.2 F1 points (Danish XLM-R 25%), well within the range expected from different tokenisation-boundary handling between the two evaluation paths. This cross-check confirms that the training pipeline reported consistent and reproducible scores throughout.

11 Discussion

11.1 What multi-seed replication revealed

The three-seed replication substantially changes the picture relative to the single-seed results, in two important ways.

Single-seed dev was misleading. Seed 42 happened to be a favorable initialisation for Norwegian XLM-R (86.49 vs 85.52). Across three seeds, English XLM-R averages marginally higher on dev (85.87 vs 85.45). Had we reported only seed 42 and drawn conclusions from dev performance, we would have overstated the Norwegian advantage.

The test set tells a clearer story. On the held-out test set, Norwegian is consistently ahead for both encoders: +1.86 F1 for XLM-R ($p = 0.10$, borderline) and +3.36 F1 for mBERT ($p = 0.046$, significant). The mBERT result is the most reliable: the gap is large, variance is low for both systems, and statistical significance is achieved even with only two usable seed pairs. The XLM-R result is directionally consistent but remains borderline; a fourth or fifth seed would likely tip it to significance.

The honest summary: Norwegian is a significantly better zero-shot transfer source than English for mBERT on the Danish test set, and a consistently (but not yet significantly) better source for XLM-R. The case is strongest on test data and weakest on dev, which reinforces the standard recommendation to report held-out test performance rather than dev performance for cross-lingual comparisons.

11.2 Inconvenient finding: EN mBERT best fine-tuned, DA XLM-R < DA BERT

Two results in Table 5 cut against the paper’s preferred narrative and deserve direct attention.

English mBERT fine-tuned on Danish is the best transfer-plus-fine-tuning model in the whole study (89.4 dev, 85.5 test), outperforming every Norwegian transfer variant. This directly contradicts a naive reading of the linguistic-distance hypothesis: if proximity predicts transfer quality, the Norwegian fine-tuned models should rank highest. The likely explanation is that mBERT trained on English benefits from *more English training data* (English has 20,142 sentences vs Norwegian’s 4,183), which gives it a richer representation before fine-tuning, and that this data-volume advantage outweighs the typological-distance advantage of Norwegian. It also reflects the fact that EN→DA

mBERT FT starts from a checkpoint trained on five times as much labeled data. The distance hypothesis operates most cleanly in the *zero-shot* regime, where data volume cannot compensate. Once fine-tuning on Danish is introduced, the data-quality and initialisation effects dominate.

Danish XLM-R underperforms Danish BERT on Danish-only training (88.6 vs 89.0 dev, 84.9 vs 85.3 test). This is unexpected given XLM-R’s generally stronger cross-lingual results. A plausible explanation is that `xlm-roberta-base` allocates vocabulary and attention across 100 languages and may be slightly undertrained on Danish relative to `bert-base-multilingual-cased`, which uses a shared vocabulary more efficiently for smaller language sets. Another possibility is that the difference is within run-to-run noise, and a multi-seed experiment would show no consistent gap. Either way, the result weakens the blanket claim that XLM-R is the strongest encoder.

11.3 Encoder choice

XLM-R leads on source-language baselines (92.0 vs 91.2 for Norwegian, 81.7 vs 80.1 for English) and on zero-shot transfer (86.5 vs 83.4 for Norwegian zero-shot). The XLM-R advantage is most consistent in zero-shot settings; it narrows or reverses after fine-tuning on Danish. This suggests that XLM-R’s broader coverage helps when the model must generalise without target-language data, but mBERT’s more concentrated vocabulary can be competitive once Danish labels are available.

11.4 Data efficiency

The Danish learning curve (Figure 2) shows that transfer is useful even before any Danish supervision is added, but a modest amount of target-language annotation quickly recovers most of the full-data performance. The Danish XLM-R model reaches the English zero-shot reference at around 50% of the training data and approaches the Norwegian zero-shot reference at 50–100%. In practical terms: for a project with a limited annotation budget, starting with a Norwegian zero-shot model and labeling ~50% of the Danish training set is a reasonable cost-effective strategy.

11.5 Limitations

The main limitations are: (1) single-seed experiments, meaning effect sizes are uncertain; (2) only English and Norwegian as source languages, while

Swedish or German experiments would strengthen the distance-gradient claim; (3) evaluation on one NER benchmark family, so results may not generalise to other Danish NER datasets or domain-shifted text; (4) no hyperparameter tuning per source language, which may disadvantage some systems.

12 Conclusion

This report presents a systematic, multi-seed study of cross-lingual NER transfer to Danish, comparing English and Norwegian as source languages across mBERT and XLM-R. Experiments are evaluated with strict, unlabeled, and loose span F1 on both development and held-out test sets. The headline comparison is replicated across three random seeds and assessed with paired *t*-tests.

Main finding. Norwegian is a statistically significantly better zero-shot transfer source than English for mBERT on the Danish test set ($\Delta = +3.36$, $p = 0.046$). For XLM-R the advantage is consistent but borderline ($\Delta = +1.86$, $p = 0.10$). Critically, the single-seed development result that originally motivated the claim (NO=86.49 vs EN=85.52, seed 42) was a favorable outlier for Norwegian: averaged across three seeds, English XLM-R is marginally ahead on dev. This makes the test-set replication especially important — it is the evaluation that holds up under multi-seed scrutiny.

Why Norwegian transfers better. Norwegian is $5.7\times$ closer to Danish than English in WALS syntactic feature space (0.021 vs 0.119 cosine distance). This distance gap is the largest in the five-language analysis (NO < SV < EN $\approx$ DE), consistent with the genealogical structure of Germanic. The transfer advantage is strongest in the zero-shot regime where source-language choice cannot be compensated by target-language fine-tuning.

When English wins. The best single fine-tuned transfer result is English mBERT + Danish fine-tuning (89.4 dev, 85.5 test), outperforming all Norwegian transfer variants. English has five times more training data than Norwegian, and this volume advantage outweighs typological distance once Danish labels are available. The distance hypothesis applies cleanly only in the zero-shot setting.

Entity-type and metric analysis. ORG is the hardest category and shows the most cross-system

variation (74.9–83.5 dev strict F1); PER is most consistent (89.6–93.7). MISC scores are zero for all systems due to a data-format artifact: MISC labels reside in a secondary annotation column not read by the evaluation script. Unlabeled and loose F1 are consistently 3–5 points above strict, with the gap largest for low-resource Danish training (10%), where models detect spans but mislabel them more often.

Practical recommendation. For zero-shot Danish NER, use Norwegian as the source language — it is both the linguistically motivated choice and the empirically stronger one. If any Danish annotation budget is available, fine-tuning an English mBERT checkpoint on Danish is a competitive and readily available baseline. Full Danish supervision (89.0 BERT, 88.6 XLM-R dev strict F1) remains the ceiling and should be the target when annotation resources permit.

Next steps. The two most impactful extensions are: (1) running XLM-R with one or two additional seeds to confirm or rule out significance of the +1.86 test gap; (2) adding Swedish transfer experiments to test whether the distance gradient ($NO < SV < EN$) predicts a corresponding F1 gradient, which would turn two data points into a testable trend.

References

- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Paco Guzman, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.
- Kenneth Enevoldsen, Lasse Hansen, and Kristoffer L. Nielbo. 2021. [DaCy: A unified streamlined Danish NLP framework](#). In *Proceedings of the 23rd Nordic Conference on Computational Linguistics (NoDaLiDa)*, pages 206–216.
- Yingwen Fu, Nankai Lin, Boyu Chen, Ziyu Yang, and Shengyi Jiang. 2023. [Cross-lingual named entity recognition for heterogenous languages](#). *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 31:371–382.
- Rasmus Hvingelby, Amalie Brogaard Pauli, Maria Barrett, Christina Rosted, Lasse Malm Lidegaard, and Anders Søgaard. 2020. [DaN+: Danish nested named entities and lexical normalization](#). In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6649–6662.
- Aditya Khan, Mason Shipton, David Anugraha, Kaiyao Duan, Phuong H. Hoang, Eric Khui, A. Seza Dogruoz, and En-Shiun Annie Lee. 2025. [URIEL+: Enhancing linguistic inclusion and usability in a typological and multilingual knowledge base](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 6937–6952.
- Patrick Littell, David R. Mortensen, Ke Lin, Katherine Kairis, Carlisle Turner, and Lori Levin. 2017. [URIEL and lang2vec: Representing languages as typological, geographical, and phylogenetic vectors](#). In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 8–14.
- Stephen Mayhew, Terra Blevins, Shuheng Liu, Marek Suppa, Hila Gonen, Joseph Marvin Imperial, Borje F. Karlsson, Peiqin Lin, Nikola Ljubešić, LJ Miranda, Barbara Plank, Arij Riab, and Yuval Pinter. 2024. [Universal NER: A gold-standard multilingual named entity recognition benchmark](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*.
- Barbara Plank. 2020. [Neural cross-lingual transfer and limited annotated data for named entity recognition in Danish](#). In *Proceedings of the 23rd Nordic Conference on Computational Linguistics (NoDaLiDa)*, pages 344–349.
- Jimin Sohn, Haeji Jung, Alex Cheng, Joeon Kang, Yilin Du, and David R. Mortensen. 2024. [Zero-shot cross-lingual NER using phonemic representations for low-resource languages](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Rob Van Der Goot, Esther Ploeger, Verena Blaschke, and Tanja Samardžić. 2025. [DistaLs: a comprehensive collection of language distance measures](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 307–318.